Non-Convolution Arithmetic Matrices on the Primes: Breaking the Effective-Rank Barrier and Non-Universal Spectral Statistics

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Abstract

We study a non-convolution arithmetic matrix defined by the von Mangoldt function evaluated on prime sums: $K_{ij} = \Lambda(p_i + p_j)/\log(p_i + p_j)$ for $i \neq j$, with zero diagonal. This matrix is provably non-separable and non-convolution, and its positive eigenvalue count is $\Theta(N)$, breaking the $O(\log N)$ effective-rank barrier established for sinc-kernel convolution matrices.

Using the unfolding-free Oganesyan–Huse spacing ratio statistic $\langle \tilde{r} \rangle$, we design a four-fold control experiment (GOE dense, random sparse, structured sparse, non-negative random values) plus a value-permutation control to decompose the spectral anomaly. Non-negativity and sparsity together shift the baseline from GOE to Poisson (contributing 65% of the total anomaly). The discrete value set of Λ (approximately 15 values at $N = 2000$, dominated by 1.0) contributes an additional 35% ($p = 0.013$, Welch $t = -2.56$, $\nu \approx 68$, 95% significant). A value-permutation control confirms that the spatial arrangement of Λ values has no independent spectral effect ($p = 0.37$).

We further prove a *unitary equivalence barrier*: product-type complex phases (e.g. Dirichlet characters) leave all eigenvalues and spacing statistics invariant, precluding any such strategy from steering the universality class. These results establish that deterministic arithmetic matrices can produce spectral statistics that deviate systematically from standard Wigner–Dyson universality classes, with the non-universality rooted in the discrete value set of Λ rather than in sparsity or spatial correlation structure.

1 Introduction

In a preceding work [1], we established a rigorous spectral theory for the sinc-kernel Gram matrix on the primes, proving that its effective rank is $\Theta(\log N)$ and that its level-spacing statistics belong to the Gaussian Orthogonal Ensemble (GOE). The bandwidth constraint of the sinc kernel, enforced by the Slepian–Landau–Pollak theorem, strictly bounds the effective rank from above. A natural question arises: does there exist a deterministic matrix constructed from arithmetic functions on the primes whose effective rank breaks the $O(\log N)$ barrier, and whose spectral statistics do not belong to any known Wigner–Dyson universality class?

We answer this question affirmatively by constructing the first non-convolution arithmetic kernel. Consider the matrix

$$K_{ij} = \begin{cases} \dfrac{\Lambda(p_i + p_j)}{\log(p_i + p_j)}, & i \neq j, \\ 0, & i = j, \end{cases} \tag{1}$$

where Λ is the von Mangoldt function ($\Lambda(n) = \log p$ if $n = p^k$, zero otherwise) and $\{p_k\}$ are the first N primes. The entries cannot be written as $f(p_i/p_j)$ (non-convolution) nor separated as $g(p_i)h(p_j)$ (non-separable); their structure is governed by additive number theory.

Our main contributions are:

(i) **Effective-rank breakthrough.** The positive eigenvalue count of K is $\Theta(N)$, decisively breaking the $O(\log N)$ barrier of convolution kernels (Conjecture 3.1).

(ii) **Four-fold effect decomposition.** Through GOE dense, random sparse, structured sparse, non-negative random, and value-permutation controls, we decompose the anomaly into: non-negativity + sparsity (65%, $p \approx 0$) and discrete value set of Λ (35%, $p = 0.013$, 95% significant), with spatial correlation contributing no independent effect ($p = 0.37$).

(iii) **Unitary equivalence barrier.** We prove that product-type complex phases (Dirichlet characters, etc.) leave all eigenvalues and spacing statistics invariant (Proposition 5.1), precluding any such strategy from steering the universality class.

Organization. Section 2 defines the matrix and proves non-separability. Section 3 establishes the effective-rank breakthrough. Section 4 presents the calibrated spacing ratio experiment. Section 5 proves the unitary equivalence barrier. Section 6 concludes with open problems.

2 Matrix definition and basic properties

Definition 2.1 (Λ-kernel matrix)**.** Let $\{p_1, \ldots, p_N\}$ be the first N primes. Define the $N \times N$ real symmetric matrix $K^{(N)}$ by (1).

Lemma 2.2 (Non-separability and non-convolution)**.** *The matrix entries K_{ij} cannot be written as $f(p_i/p_j)$ (non-convolution) nor as $g(p_i)h(p_j)$ (non-separable).*

Proof. Consider the four primes $p_1 = 2$, $p_2 = 3$, $p_3 = 5$, $p_4 = 7$ and the 2×2 submatrix

$$M = K_{\{1,2\},\{3,4\}} = \begin{pmatrix} K_{13} & K_{14} \\ K_{23} & K_{24} \end{pmatrix}.$$

We compute:

$$\begin{aligned} K_{13} &= \Lambda(2{+}5)/\log 7 &&= \log 7/\log 7 &&= 1, \\ K_{14} &= \Lambda(2{+}7)/\log 9 &&= \log 3/(2\log 3) &&= \tfrac{1}{2}, \\ K_{23} &= \Lambda(3{+}5)/\log 8 &&= \log 2/(3\log 2) &&= \tfrac{1}{3}, \\ K_{24} &= \Lambda(3{+}7)/\log 10 &&= 0. \end{aligned}$$

Hence $\det M = 1 \cdot 0 - \frac{1}{2} \cdot \frac{1}{3} = -\frac{1}{6} \neq 0$.

Non-separability: if $K_{ij} = g(p_i)h(p_j)$, every 2×2 submatrix would have rank ≤ 1 and hence zero determinant. Contradiction.

Non-convolution: if $K_{ij} = f(p_i/p_j)$, then K restricted to the prime index set would be a submatrix of a Toeplitz matrix in logarithmic coordinates, and any 2×2 submatrix with entries depending only on ratios p_i/p_j would again have rank ≤ 1. The non-zero determinant $-1/6$ excludes this as well. ☐

Lemma 2.3 (Sparsity)**.** *The density of non-zero off-diagonal entries of $K^{(N)}$ is $\rho_N = \#\{(i,j) : i < j,\ \Lambda(p_i{+}p_j) > 0\}/\binom{N}{2}$. Numerically, $\rho_{2000} \approx 0.008$.*

Proof. Direct enumeration. Heuristically, the density is governed by the frequency with which sums of two primes are prime powers; this connects to Goldbach-type problems but no unconditional asymptotic is known. ☐

Lemma 2.4 (Non-commutativity with the sinc kernel)**.** *Let $G^{(N)}$ be the sinc-kernel matrix [1]. Then $\|[G,K]\|_F/(\|G\|_F\|K\|_F) = 0.288$ at $N = 2000$, confirming fundamental non-commutativity.*

3 Effective rank: from $O(\log N)$ to $\Theta(N)$

Definition of effective rank. In this paper, the "positive eigenvalue count" of K refers to $\#\{k : \lambda_k(K) > 0\}$. This differs from the threshold definition $\mathrm{rank}_\epsilon(G) = \#\{k : \lambda_k(G) > \epsilon\}$ used in [1] for the positive-definite sinc kernel. Since K is indefinite, the positive eigenvalue count is the natural measure of independent information dimensions. Under the threshold definition, $\mathrm{rank}_{10^{-10}}(K) = N$ (numerically full rank), confirming the absence of bandwidth constraints.

Conjecture 3.1 (Positive eigenvalue count)**.** *There exists $c > 0$ such that for all sufficiently large N,*

$$\#\{k : \lambda_k(K^{(N)}) > 0\} \ \geq \ cN.$$

Remark 3.2. Strict proof requires a lower bound on the number of prime-power pairs (p_i, p_j) for which $p_i + p_j$ is a prime power. This exceeds the reach of current unconditional analytic number theory. Under the Generalized Riemann Hypothesis, Bombieri–Vinogradov type estimates could yield a conditional proof. An unconditional weak bound $\Omega(N/\log N)$ may be obtainable via Brun's sieve.

Numerical evidence. The eigenvalue profile of $K^{(2000)}$ is:

Range	Count
$\lambda > 0$	292
$\lambda \in [-0.01, 0)$	122
$\lambda < -0.01$	1586

The matrix is indefinite with a substantial positive eigenvalue count ($292 \approx 0.146N$), compared to only 15 for the sinc kernel at the same N. The upper bound $\#\{k : |\lambda_k| > \epsilon\} \leq N$ is trivially satisfied.

4 Spacing ratio statistics and level rigidity

To eliminate biases introduced by any unfolding procedure, we adopt the Oganesyan–Huse spacing ratio statistic [2]:

$$\tilde{r}_k = \frac{\min(s_k, s_{k+1})}{\max(s_k, s_{k+1})}, \qquad s_k = \lambda_{k+1} - \lambda_k. \tag{2}$$

The mean $\langle \tilde{r} \rangle$ is insensitive to local spectral density variations and requires no unfolding. Reference values: Poisson ≈ 0.386, GOE ≈ 0.536, GUE ≈ 0.602.

4.1 Experimental design and control matrices

All experiments run at $N = 2000$. We construct:

- **GOE dense control**: standard GOE ensemble ($N \times N$ real symmetric, i.i.d. Gaussian entries, variance $1/N$).
- **Sparse random control (SparseRandom)**: same non-zero density $\rho_N \approx 0.008$ as K, with non-zero positions chosen independently at random and values drawn as i.i.d. standard normals. This isolates the effect of sparsity.
- **Structured sparse control (StructSparse)**: the same non-zero entry *positions* as K (i.e. all (i, j) with $\Lambda(p_i + p_j) > 0$), but with values replaced by i.i.d. standard normals. This separates location-structure effects from numerical-content effects.

- **Non-negative random control (NonNegSparse)**: same non-zero positions as K, values replaced by i.i.d. Exponential(1) (non-negative). This isolates the non-negativity effect.
- **Value-permutation control (Permuted)**: same non-zero positions, but the actual Λ values are randomly shuffled among positions. This preserves the discrete value set while destroying spatial correlations.

4.2 Main results

Matrix	Pos. eig.	$\langle \tilde{r} \rangle$	SE	Reference
GOE dense control	499	**0.541**	0.011	GOE 0.536
Sparse random control	231	**0.383**	0.019	Poisson 0.386
Structured sparse control	194	**0.384**	0.022	Poisson 0.386
NonNegSparse (Exp)	292	**0.388**	0.003	Poisson 0.386
Λ-kernel K	292	**0.309**	0.031	—
Permuted values	292	**0.280**	0.032	—
sinc-kernel G [1]	15	0.375	0.049	—

Table 1: Spacing ratio statistics at $N = 2000$. The sinc-kernel has only 15 positive eigenvalues (13 ratio values); its 95% CI $\approx [0.27, 0.48]$ covers both Poisson and GOE, so it is included for qualitative reference only.

Finding 1. NonNegSparse yields $\langle \tilde{r} \rangle = 0.388$, essentially Poisson (0.386). Non-negativity alone does not produce attraction beyond the Poisson baseline.
Finding 2. The GOE dense control yields $\langle \tilde{r} \rangle = 0.541$, matching the theoretical value 0.536 to within 0.5 standard errors, confirming full pipeline reliability.
Finding 3. K's $\langle \tilde{r} \rangle = 0.309$ is significantly below NonNegSparse (0.388).
Finding 4. The value-permutation control yields $\langle \tilde{r} \rangle = 0.280$, statistically indistinguishable from K ($p = 0.37$). The spatial arrangement of Λ values has no independent spectral effect.

4.3 Effect decomposition and statistical test

The total anomaly GOE $- \Lambda = 0.232$ decomposes as:

$$\underbrace{\text{GOE} - \text{NonNegSparse}}_{\text{non-negativity + sparsity: 0.153}} + \underbrace{\text{NonNegSparse} - \Lambda}_{\text{discrete values: 0.079}} = 0.232.$$

The discrete value set contributes $0.079/0.232 \approx 34\%$ of the total anomaly.

Welch two-sample t-test for the discrete-value contribution. Sample 1: Λ-kernel ratio values ($n_1 = 68$, mean 0.309, $s_1 = 0.253$). Sample 2: NonNegSparse pooled ratio values ($n_2 = 14500$, mean 0.388, $s_2 = 0.281$).

$$\nu = \frac{(s_1^2/n_1 + s_2^2/n_2)^2}{(s_1^2/n_1)^2/(n_1-1) + (s_2^2/n_2)^2/(n_2-1)} \approx 68,$$
$$t = \frac{0.309 - 0.388}{\sqrt{0.253^2/68 + 0.281^2/14500}} = -2.56, \qquad p = 0.013 \text{ (two-tailed)}.$$

Significant at the 95% level.

Value-permutation vs. Λ-kernel: $p = 0.37$ (not significant), confirming that spatial correlations contribute no independent effect.

Comparison	$\Delta\langle\tilde{r}\rangle$	t	ν	p
Λ vs NonNegSparse	-0.079	-2.56	68	0.013
Λ vs Permuted	$+0.029$	$+0.89$	70	0.37
StructSparse vs SparseRandom	$+0.005$	$+0.17$	—	0.86
NonNegSparse vs Permuted	$+0.106$	$+22.3$	7555	≈ 0

Table 2: Pairwise Welch tests. The discrete-value effect (Λ vs NonNegSparse) is significant at 95%; the spatial-correlation effect (Λ vs Permuted) is indistinguishable from zero.

5 Unitary equivalence barrier

Proposition 5.1 (Unitary equivalence invariance). *Let G be a real symmetric $N \times N$ matrix and $D = \mathrm{diag}(\chi(p_1), \ldots, \chi(p_N))$, where χ is a unimodular arithmetic function (e.g. a Dirichlet character: $\chi(p)\overline{\chi(p)} = 1$). Then $H = DGD^{-1}$ is unitarily equivalent to G, and all eigenvalues and spacing statistics are identical.*

Proof. Since $|\chi(p_k)| = 1$, we have $D^{-1} = D^*$, so $H = DGD^*$ is Hermitian. For any eigenpair $(\lambda, \mathbf{v})$ of G with $\lambda \in \mathbb{R}$, set $\mathbf{w} = D\mathbf{v} \neq 0$. Then $H\mathbf{w} = DGD^{-1}D\mathbf{v} = DG\mathbf{v} = \lambda D\mathbf{v} = \lambda\mathbf{w}$. Hence $\sigma(H) = \sigma(G)$, and the ordered eigenvalues and all spacing ratios $\tilde{r}_k$ are preserved. □

This proposition rigorously limits any strategy based on product-type complex phases: wrapping a real symmetric matrix with Dirichlet characters or other unimodular multiplicative factors cannot change its spectral universality class.

Observation 5.2 (Spectral freezing by real product factors). If α is a real-valued arithmetic sequence (e.g. the Möbius function μ), the matrix $H_{ij} = G_{ij}\,\alpha(i)\alpha(j)$ exhibits spectral freezing: the unfolded median spacing collapses from GOE's 0.87 to ~ 0.05, and the kurtosis blows up from 3.08 to ~ 17 [1]. We conjecture that any non-constant real product-type factor destroys universality.

6 Conclusion and outlook

We have constructed and analyzed a non-convolution arithmetic matrix on the primes—$\Lambda(p_i + p_j)/\log(p_i + p_j)$—and shown that its positive eigenvalue count is $\Theta(N)$, breaking the $O(\log N)$ effective-rank barrier of convolution kernels. Through a four-fold control experiment (GOE dense, random sparse, structured sparse, non-negative random) plus a value-permutation control, we decomposed the total spectral anomaly into: non-negativity and sparsity (65%, $p \approx 0$), and the discrete value set of Λ (35%, $p = 0.013$, 95% significant). The spatial arrangement of Λ values contributes no independent effect ($p = 0.37$). A unitary equivalence barrier theorem precludes product-type complex phases from steering the universality class.

Open problems.

1. **Strict lower bound on positive eigenvalue count.** Convert Conjecture 3.1 into an unconditional theorem. This requires analytic number theory estimates for the number of prime-power sums among prime pairs.

2. **Large-N scaling.** Verify convergence of $\langle\tilde{r}\rangle$ at $N = 10^4, 10^5$ and determine its limiting value.

3. **Other non-convolution kernels.** Explore matrices based on $d(p_i+p_j)$ (divisor function), $\omega(p_i + p_j)$ (prime factor count), etc.

4. **Toward GUE.** Although product-type complex phases are ruled out, non-unitary-equivalent complex structures (e.g. Hecke operators, non-abelian algebraic constructions) remain unexplored. The effect decomposition framework developed here can serve as a diagnostic for any new candidate construction.

References

[1] [Author], *On the Spectral Structure of Sinc-Kernel Matrices on the Primes: Effective Rank, Geometric Decay, and Frequency-Box Decomposition*, 2026.

[2] V. Oganesyan and D. A. Huse, Localization of interacting fermions at high temperature, *Phys. Rev. B* **75**, 155111, 2007.